

## STUTTGART, AR, USA

WMO: 723416

Lat: 34.600N

Lon: 91.574W

Elev: 68

StdP: 100.51

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13925

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -6.2       | -3.8 | -13.7                         | 1.2 | -4.1 | -11.2 | 1.4 | -2.1 | 12.0                  | 8.2  | 11.1 | 7.8  | 4.9                   | 20   | 0.478 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.7                    | 36.2            | 25.5 | 34.8 | 25.3 | 33.7 | 25.0 | 27.1                | 33.3 | 26.5 | 33.1 | 25.9 | 32.0 | 3.6                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 25.3                            | 20.6 | 30.5 | 24.6 | 19.7 | 29.8 | 24.0 | 19.0 | 29.3 | 85.5          | 34.0 | 82.6 | 33.2 | 80.1 | 32.3 | 29.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 10.8          | 9.3  | 8.4 | -8.9                       | 37.9 | 2.5                | 2.0 | -10.7                                              | 39.4 | -12.2      | 40.6 | -13.6      | 41.7 | -15.4      | 43.2 |    |    |
| (5)               |      |     | -10.5                      | 28.0 | 2.3                | 0.9 | -12.2                                              | 28.6 | -13.5      | 29.1 | -14.8      | 29.6 | -16.5      | 30.2 |    |    |

## Monthly Climatic Design Conditions

|       |                                                                                  | Annual                                                                           | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|-------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|       |                                                                                  | (d)                                                                              | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)   | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg                                                                            | 17.3   | 5.4   | 7.8   | 12.1  | 17.3  | 22.3  | 26.6  | 27.9  | 27.4  | 23.6  | 17.5  | 11.3  | 7.1  | (6)  |
| (7)   |                                                                                  | DBStd                                                                            | 9.11   | 5.65  | 5.69  | 5.39  | 4.41  | 3.78  | 2.58  | 2.24  | 2.44  | 3.50  | 4.67  | 4.83  | 5.16 | (7)  |
| (8)   |                                                                                  | HDD10.0                                                                          | 469    | 162   | 101   | 40    | 3     | 0     | 0     | 0     | 0     | 0     | 2     | 40    | 121  | (8)  |
| (9)   |                                                                                  | HDD18.3                                                                          | 1625   | 400   | 297   | 203   | 70    | 12    | 0     | 0     | 0     | 4     | 74    | 214   | 349  | (9)  |
| (10)  |                                                                                  | CDD10.0                                                                          | 3118   | 21    | 38    | 104   | 223   | 382   | 499   | 556   | 540   | 409   | 235   | 80    | 31   | (10) |
| (11)  |                                                                                  | CDD18.3                                                                          | 1232   | 1     | 2     | 9     | 39    | 136   | 249   | 298   | 282   | 163   | 48    | 4     | 1    | (11) |
| (12)  |                                                                                  | CDH23.3                                                                          | 12531  | 1     | 5     | 52    | 286   | 1121  | 2642  | 3403  | 3063  | 1492  | 438   | 27    | 1    | (12) |
| (13)  |                                                                                  | CDH26.7                                                                          | 5528   | 0     | 0     | 4     | 54    | 388   | 1209  | 1637  | 1482  | 617   | 134   | 3     | 0    | (13) |
| (14)  | Wind                                                                             | WSAvg                                                                            | 3.9    | 4.5   | 4.6   | 4.8   | 4.7   | 4.2   | 3.5   | 3.0   | 2.8   | 3.2   | 3.6   | 4.1   | 4.2  | (14) |
| (15)  | Precipitation                                                                    | PrecAvg                                                                          | 1226   | 99    | 96    | 122   | 129   | 119   | 81    | 89    | 67    | 80    | 115   | 104   | 122  | (15) |
| (16)  |                                                                                  | PrecMax                                                                          | 1995   | 184   | 212   | 281   | 331   | 320   | 146   | 222   | 157   | 170   | 357   | 260   | 264  | (16) |
| (17)  |                                                                                  | PrecMin                                                                          | 908    | 7     | 33    | 43    | 39    | 21    | 34    | 18    | 15    | 2     | 1     | 23    | 16   | (17) |
| (18)  |                                                                                  | PrecStd                                                                          | 265    | 48    | 43    | 58    | 76    | 59    | 38    | 53    | 35    | 46    | 84    | 56    | 58   | (18) |
| (19)  | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                             | DB     | 22.3  | 23.8  | 27.4  | 30.1  | 33.6  | 36.3  | 37.4  | 37.7  | 35.9  | 32.4  | 27.0  | 22.5 | (19) |
| MCWB  |                                                                                  |                                                                                  | 17.7   | 18.3  | 19.0  | 21.0  | 22.9  | 24.7  | 25.9  | 25.9  | 24.6  | 22.1  | 20.2  | 18.5  | (20) |      |
| 2%    |                                                                                  | DB                                                                               | 18.9   | 21.3  | 24.8  | 27.9  | 32.2  | 35.0  | 36.1  | 36.2  | 33.6  | 30.0  | 23.0  | 19.2  | (21) |      |
|       |                                                                                  | MCWB                                                                             | 16.1   | 16.9  | 17.6  | 19.8  | 22.6  | 24.9  | 25.9  | 25.8  | 24.2  | 21.7  | 17.2  | 16.4  | (22) |      |
| 5%    |                                                                                  | DB                                                                               | 17.2   | 18.9  | 22.6  | 27.0  | 30.9  | 33.7  | 34.1  | 34.2  | 32.3  | 27.8  | 21.3  | 17.4  | (23) |      |
|       |                                                                                  | MCWB                                                                             | 14.1   | 15.3  | 16.5  | 19.2  | 22.2  | 24.4  | 25.6  | 25.5  | 23.8  | 19.7  | 16.9  | 14.6  | (24) |      |
| 10%   |                                                                                  | DB                                                                               | 14.1   | 17.1  | 20.9  | 24.9  | 28.8  | 32.5  | 33.0  | 32.9  | 30.9  | 26.0  | 18.9  | 15.1  | (25) |      |
|       |                                                                                  | MCWB                                                                             | 11.0   | 13.4  | 15.5  | 18.3  | 21.7  | 24.1  | 25.3  | 25.1  | 23.1  | 19.0  | 15.6  | 12.3  | (26) |      |
| (27)  |                                                                                  | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 18.8  | 19.5  | 20.5  | 22.8  | 25.0  | 27.0  | 27.9  | 26.2  | 24.5  | 21.3  | 19.1 | (27) |
| MCD B |                                                                                  |                                                                                  |        | 21.2  | 22.3  | 25.4  | 28.2  | 30.4  | 33.3  | 33.4  | 34.1  | 33.2  | 29.8  | 24.9  | 21.7 | (28) |
| 2%    | WB                                                                               |                                                                                  | 16.7   | 17.8  | 18.9  | 21.4  | 24.1  | 26.1  | 27.1  | 27.2  | 25.3  | 23.2  | 19.1  | 17.3  | (29) |      |
|       | MCD B                                                                            |                                                                                  | 18.4   | 20.3  | 23.4  | 26.5  | 29.2  | 32.7  | 33.1  | 33.6  | 31.4  | 27.9  | 21.9  | 18.8  | (30) |      |
| 5%    | WB                                                                               |                                                                                  | 14.6   | 15.9  | 17.5  | 20.4  | 23.4  | 25.5  | 26.5  | 26.5  | 24.6  | 21.7  | 17.7  | 15.1  | (31) |      |
|       | MCD B                                                                            |                                                                                  | 16.8   | 18.5  | 21.1  | 25.1  | 28.5  | 31.6  | 32.9  | 33.2  | 30.2  | 25.4  | 20.4  | 17.2  | (32) |      |
| 10%   | WB                                                                               |                                                                                  | 11.7   | 13.7  | 16.0  | 19.3  | 22.5  | 24.9  | 26.0  | 25.8  | 23.9  | 20.3  | 15.7  | 12.8  | (33) |      |
|       | MCD B                                                                            |                                                                                  | 13.9   | 16.4  | 19.6  | 23.5  | 27.6  | 30.6  | 32.1  | 31.9  | 28.8  | 24.2  | 18.6  | 14.9  | (34) |      |
| (35)  | Mean Daily<br>Temperature<br>Range                                               |                                                                                  | MDBR   | 8.5   | 9.2   | 9.9   | 10.6  | 10.0  | 9.8   | 9.7   | 10.3  | 11.2  | 11.9  | 9.9   | 8.4  | (35) |
| 5% DB |                                                                                  |                                                                                  | MCD BR | 11.6  | 12.1  | 11.9  | 12.1  | 11.5  | 10.8  | 10.6  | 11.6  | 12.4  | 13.9  | 12.3  | 10.8 | (36) |
|       |                                                                                  | MCWBR                                                                            | 9.1    | 9.0   | 7.5   | 6.2   | 4.7   | 4.1   | 4.1   | 4.4   | 5.0   | 6.5   | 7.9   | 8.6   | (37) |      |
|       |                                                                                  | MCD BR                                                                           | 10.2   | 11.0  | 10.7  | 10.4  | 9.9   | 9.8   | 9.9   | 10.5  | 10.6  | 11.1  | 10.5  | 9.7   | (38) |      |
| 5% WB |                                                                                  | MCWBR                                                                            | 9.1    | 8.9   | 7.5   | 6.5   | 4.8   | 4.3   | 4.5   | 4.6   | 4.7   | 6.3   | 8.3   | 8.5   | (39) |      |
|       |                                                                                  |                                                                                  |        |       |       |       |       |       |       |       |       |       |       |       |      | (39) |
| (40)  | Clear-Sky<br>Solar<br>Irradiance                                                 | taub                                                                             | 0.323  | 0.332 | 0.362 | 0.405 | 0.436 | 0.466 | 0.495 | 0.486 | 0.419 | 0.360 | 0.336 | 0.320 | (40) |      |
| (41)  |                                                                                  | taud                                                                             | 2.487  | 2.455 | 2.397 | 2.273 | 2.247 | 2.177 | 2.129 | 2.140 | 2.316 | 2.460 | 2.483 | 2.507 | (41) |      |
| (42)  |                                                                                  | Ebn at Noon                                                                      | 883    | 914   | 910   | 881   | 855   | 825   | 798   | 799   | 840   | 871   | 860   | 866   | (42) |      |
| (43)  |                                                                                  | Edh at Noon                                                                      | 85     | 98    | 112   | 134   | 139   | 149   | 155   | 151   | 121   | 96    | 85    | 79    | (43) |      |
| (44)  | All-Sky Solar<br>Radiation                                                       | RadAvg                                                                           | 2.50   | 3.00  | 4.18  | 5.28  | 5.85  | 6.57  | 6.34  | 5.83  | 4.97  | 3.86  | 2.77  | 2.21  | (44) |      |
| (45)  |                                                                                  | RadStd                                                                           | 0.22   | 0.45  | 0.39  | 0.37  | 0.40  | 0.45  | 0.33  | 0.44  | 0.37  | 0.50  | 0.30  | 0.23  | (45) |      |

## Historical Trends

|                              |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)                          | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| (46) Station Only            | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     |         |         |
| (47) Regional (27 neighbors) | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     |         |         |

Nomenclature: See separate page